

CSIT314 Development Document

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Development Document

Project Overview (5%) - Ciara

Project Background

The recruitment process is an important part of the modern job market. Job seekers look for suitable job opportunities, and employers look for suitable candidates. However, the recruitment process is often difficult and time consuming for both candidates and employers.

Many job search platforms provide many job postings and applications, but it is not always easy to find the best match. Job seekers often spend a lot of time searching for relevant jobs, and employers receive many applications from candidates who do not meet the job requirements. This makes the recruitment process inefficient and slow.

To solve this problem, the Intelligent Talent Matching Platform is proposed. This system is designed to match candidates and job opportunities based on their profiles, skills, education, and work experience. The system aims to make the recruitment process faster and more efficient. In this system, candidates can create profiles and upload their information, and employers can create job postings with specific requirements. The system then recommends the most relevant jobs to candidates and the most suitable candidates to employers.

Problem Definition

The biggest problem for candidates in the job market is that it is often very difficult to find a suitable job that fits their skills and expectations. Also, many talented candidates with strong resumes do not even get a chance to reach the first interview because automated AI filters may reject their resumes due to keyword issues.

As for companies, they receive many irrelevant applications from candidates who apply for many jobs without meeting the requirements. With such a high number of applications, it becomes very difficult to identify the most suitable candidates. As a result, companies may hire the wrong candidate or fail to fill the position successfully.

Without an intelligent matching system with effective filtering, the process of finding a job or a suitable candidate becomes inefficient and time-consuming. Therefore, a smart recommendation system is needed in today's highly competitive job market.

System Objectives

The main objectives of the system are:

- To develop a web-based recruitment platform for candidates and employers.
- To allow candidates to create and manage their profiles.
- To allow employers to create and manage job postings.
- To provide search and filtering functions for jobs and candidate profiles.
- To implement a recommendation system for job and candidate matching.
- To improve the efficiency of the recruitment process in general.

Target Users

The target users of the system include candidates, employers, and system administrators. Candidates use the system to create profiles, search for job opportunities, and receive job recommendations. Employers use the system to post job vacancies, search for candidates, and receive candidate recommendations. System administrators manage the platform, user accounts, and system data.

Development Model (10%) - Ciara

Chosen Agile Methodology

The team at Hire Wire chose to adopt the Scrum agile methodology to assist in the development life cycle. Scrum was originally chosen because of its flexible and iterative system design, which greatly contributed to the success of the project. It made the occurrence of requirement changes more simple and easier to control without changing the scope of the project. It also accounted for stakeholder needs and adjustments to the allocation of time and effort. Additionally, the team adopted the use of Kanban board to easily track the immediate allocation of work.

Actual vs. Planned Execution

Originally, the team at Hire Wire chose to adopt 2-week sprints with the goal of keeping allocated tasks relatively short, but with enough time to complete in-depth or larger tasks when needed. This stayed true throughout the rest of the project and proved extremely effective for the effort and time allocation needed for this project. Sprint planning, weekly scrums, sprint review, and sprint retrospective meetings remained consistent throughout most of the project.

Due to some public holidays, 2 weekly scrum meetings were cancelled and rescheduled online for a shorter period of time than normal. Luckily, this did not affect the progress of the project too much and instead was a good way to summarise the work done and re-allocate a smaller amount of work, giving the team a well-deserved break.

Changes to Tasks and Tools

An unplanned change to allocated tasks that occurred was the roles and responsibilities of the team. Whilst James (planned front-end designer) was originally working on the front-end design, he was moved to backend in the middle stages of the project. This was due to the fact that the backend needed more time allocated to its development. It was also discovered that backend design effort was originally underestimated, only to be more accurately adjusted by the team later on.

A change of tools included relying more on Taiga for the initial allocation of tasks. The product owner would define features and user stories based on the previous weekly scrum and allocate appropriate effort to each task accordingly. Then, in the sprint planning, team members were allocated tasks based on a number of considerations including availability, skill set, and time. As proven in the sprint review meetings, this

worked very well for the team and allowed developers to focus on their work with minimal administrative work.

Development Progress (10%) - Qianran

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Change Management Progress (15%) - Siew Tin

Several new requirements were introduced during the development process of this project including improvements to the candidate profile, the implementation of a membership feature for both membership and non-membership users and enhancements to the search function. The search function required the system to include a function that supports returning results based on keywords, filtering, a combination of both keywords and filters as well as fuzzy search which allows typing errors while still returning related results. With the new requirements provided, our team had to revisit and make necessary adjustments to the design of the website and underlying code to ensure we meet the expected requirements.

The changes have made a significant impact on the existing design and system. Improvements to the candidate profile will require some changes to be made in the database as well as the website interface design. Furthermore, the membership feature plays a crucial role in affecting both candidate and job recommendations that will be shown to users. Membership users may receive more detailed and useful information while non-membership users will receive fewer suggestions. Enhancing the search function requires complex coding as it needs to cover all three main requirements mentioned.

Due to new changes required, the team had to regulate the project planning by revisiting the scrum backlog and reprioritising the new requirements into each sprint. New tasks were added to the sprint and all the tasks listed in the backlog were redistributed to each member in the team based on their skills and availability. There are five members in the team and each member was allocated with a task where two members focusing on the UI design, one member adding new fields to the database and two members focusing on the backend including the API logic. The tasks distributed may be switched among the team when any member is struggling or facing any challenges to reduce time spent on one specific task. Some low priority tasks were delayed guaranteeing there is ample time for implementing new features.

(left technical adjustments (must align with Technical Document))

System Analysis and Design (12%) - James

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System architecture – Nikita

The system uses a client-server architecture where the frontend communicates with the backend through REST API endpoints. The backend processes requests, manages business logic and interacts with the database.

Architecture components:

Frontend

- HTML/CSS/JavaScript
- user interaction
- profile/job pages

Backend

- Node.js + Express
- handles API requests
- processes candidate and job data

Database

- stores candidate and job information

Data flow:

User -> Frontend -> Express API -> Database -> Response to frontend

Backend Design – Nikita

The backend of the system is developed using Node.js and Express. It is responsible for handling API requests, processing job and candidate data, and connecting the frontend with the database.

Backend structure:

server.js

- starts the Express server
- connects frontend pages with backend routes
- registers API routes

routes/

- contains route files for different system features
- jobs.js handles job-related API endpoints
- candidates.js handles candidate-related API endpoints

controllers/

- contains backend logic for each feature
- jobsController.js processes job data and job search logic
- candidatesController.js processes candidate data and candidate search logic

db/

- contains database connection logic
- db.js is used to connect the backend with the database

Current API endpoints:

- GET /api/jobs
- GET /api/jobs/search
- GET /api/candidates

The backend design separates routes, controllers, and database logic. This makes the code easier to maintain and extend when new features are added, such as filtering, matching, and recommendation logic.

Meeting Records (3%) - Qianran

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